

SSTV on 11 m quick-start (sdroxide — CB/11 m)

A short, task-oriented walkthrough for sending and receiving **SSTV pictures on the 11 m (27 MHz) citizens' band** with **sdroxide**: which frequency, how to set it up, how to receive a picture and how to send one. SSTV is an analogue image mode — a picture is sent as tones, line by line, over a normal sideband — and on 11 m it is the community's picture mode, alongside the FT8 traffic. For the reference detail behind each control see [USER MANUAL.md](#) §3.6 and the 11 m band in §6.1; for the why of this fork see the [README](#).

This is the CB/SWL fork of sdroxide. SSTV itself is upstream's; what this fork adds on 11 m is the band, the per-country channel plans, the 11 m SSTV calling channels and the CB callsign in the banner.

Nederlands: [sstv-11m-quickstart.nl.md](#). Français: [sstv-11m-quickstart.fr.md](#). Italiano: [sstv-11m-quickstart.it.md](#). PDF: [sstv-11m-quickstart.en.pdf](#).

Bold names such as **SETTINGS** and **Callsign** are buttons and fields exactly as they appear on screen. Settings > Radio is a menu path.

The frequency (read this first)

- **27.700 MHz is the 11 m SSTV calling frequency** — the one the community actually uses. It sits in the “freeband” above the forty channels (the band runs to 27.860 so that channel is inside it).
 - The in-band picture channels are **27.255 (channel 23)** and **27.375 (channel 37)**.
 - **You do not have to dial any of this by hand.** Choosing a band and then **SSTV** puts the dial on that band's SSTV frequency — 27.700 on 11 m — and the **↕ FREQ** chip is there if you want a different one, or want to save your own. Part C.2 explains both.
 - Choose the mode **SSTV** — not **SSTV-FM**, which is for VHF/UHF. SSTV follows phone practice rather than the digital modes' fixed USB: **LSB on 80 and 40 m, USB on 20 m and up**, and 11 m is up, so the radio is put in **USB**.
 - **A picture is slow.** From about a minute (Robot 36) and two (Martin 1, PD120) to four and a half (Scottie DX). One station holds the frequency for that whole time — listen before you send, and move off the calling frequency to chat so others can call.
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What you need

- **An SDR or radio sdroxide supports:** RTL-SDR, RX-888, Airspy HF+, SDRplay RSP, HackRF, ELAD, PlutoSDR, a **CAT rig** over serial + sound card, TCI, OpenHPSDR or SoapySDR.
- **A 27 MHz antenna** suited to your receiver.
- **sdroxide installed** (below).
- **To send a picture you need a transceiver**, keyed by **VOX** (a sound-card rig) or **CAT**. A receive-only dongle hears and decodes pictures but cannot send them.

Installing

- **Windows** — the installer (.msi) or the portable .zip: see the [Releases page](#).
- **Linux** — the **AppImage** (one file, chmod +x and run), the .deb, or the portable tarball.
- **macOS** — the .dmg.

You can also run sdroxide as a **server** and open it in a browser: `sdroxide --server`, then `http://localhost:4950`.

Part A — One-time setup

Everything here is saved under `~/.config/sdroxide-brown/`, so you do it once.

1. Pick your radio

Open **SETTINGS** and go to the **Radio** tab. Pick your interface (for example **RTL-SDR, RX-888** or **Airspy HF+**), then the **sample rate** and the **gain**. Changes apply right after **Apply / reconnect**.

- **Linux — USB receivers:** install the bundled udev rules, or sdroxide will see the device but fail to open it: `sudo cp 60-sdroxide-*.rules /usr/lib/udev/rules.d/ && sudo udevadm control --reload`, then replug.
- **Windows — RX-888:** bind the device to **WinUSB** once with [Zadig](#) — for **both** USB ids (04B4:00F3 and 04B4:00F1).

2. Fill in your details

Go to the **General** tab:

- **Callsign** — your CB identifier, for example 26AT715. It is drawn into the picture's banner and sent as the **FSK ID**, which is how another station or a repeater reads who is sending.
- **Locator / grid** — your Maidenhead grid, used for the map and the great-circle paths; it is **not** sent on 11 m.
- **CB plan** — **World / freeband**, **CEPT/EU**, **Germany 80 channels**, **UK 27/81**, **USA** or **Australia**. It fixes the channels and the channel numbers.

3. Only to send: allow 11 m

11 m transmit is off by default — see Part C. To send a picture, switch on **Allow transmit on 11 m (CB)** on the General tab and confirm the note once. Receiving needs no switch.

Part B — Receiving a picture

1. Press **11M** on the band bar.
2. Open the **Band / Mode** popup and choose **SSTV** from the **DIGITAL** row. The SSTV panel appears under the waterfall: a **RECEIVED** gallery on the left, a transmit compositor on the right and a row of mode buttons on top.
3. Tune to **27.700 MHz** (or 27.255 / 27.375). The band buttons land on the band's SSTV calling frequencies, so the 11M band's SSTV dials take you there.
4. An incoming picture builds up **scanline by scanline** in the **LIVE** view as it arrives, then drops into the **RECEIVED** gallery, newest first.
5. **Auto** (the default) reads the mode from the VIS header — or the sync cadence if you tuned in mid-picture — so you do not have to pick one. The **Signal** meter shows the receive audio level, so you can confirm audio is reaching the decoder. If a header was misread as a slow mode, **Restart RX** drops the half-picture and starts hunting again.
6. Received pictures are saved as PNG under `~/.config/sdroxide-brown/sstv_rx/` and reload into the gallery next time.

Part C — Sending a picture

Sending is the half of SSTV that most people never try, because it needs a transmitter. This Part is the whole of it: what has to be true first, where to send, what to send, and what the rules are.

C1. What has to be true first

Four things, and the program checks every one of them for you:

- **A transceiver, keyed by VOX or CAT.** A receive-only dongle — most RTL-SDR, RX-888 and Airspy HF+ setups — decodes pictures beautifully but cannot send them. With one, **TX** is greyed and says so: *“This radio can only receive — it has no transmitter to key.”* Everything else still works, so you can load and compose a picture and send it later.
- **SWL mode off.** In SWL mode the whole **TRANSMIT** half is not drawn.
- **Allow transmit on 11 m (CB)** switched on (Part A.3). 11 m is not an amateur band, and the program’s amateur-band lockout is generic, so CB is opted into rather than assumed. The first time you switch it on, a note asks you to confirm you know what your country allows on 27 MHz; it is remembered afterwards, so it is not asked on every session.
- **A frequency inside the band.** See C2 — this part takes care of itself.

Your **callsign** is **not required** to transmit. Without one, the **FSK ID** is simply not sent and the banner’s {call} comes out empty. Set it in **Settings > General > Station** anyway — it is how another station, or a repeater, reads who is sending.

C2. Where to send — the FREQ chip

The ↕ **FREQ** chip is at the top left of the SSTV panel, and it is the one control that says where you are.

- **Choosing a band and then SSTV puts you on that band’s SSTV frequency automatically.** On 11 m that is **27.700 MHz**, the community’s picture frequency. The 11M band’s SSTV buttons take you there too, so usually there is nothing to do.
- If you were **already listening on one of the SSTV frequencies, you are left exactly where you were.** Nothing moves a dial you have deliberately placed — this only ever rescues a dial that is nowhere useful.
- Press ↕ **FREQ** to see the list. It shows the calling frequency plus the alternates, and **picking one tunes the dial.** On 11 m that is **27.700** (plain, the primary), **27.255** (channel 23) and **27.375** (channel 37).
- The chip doubles as a readout: it shows ↕ 27.700 when the dial is sitting on one of them and plain ↕ **FREQ** when it is not, so you can see at a glance whether you are where SSTV is expected.
- You can **save your own frequency** to the same list, and delete it again — which is how you keep a frequency a local group has settled on.

C3. What to send

- The **TRANSMIT** side has **five slots** that work like tabs. **Load image...**, or double-click a slot, picks a **PNG or JPEG** (16 MB limit). It is centre-cropped and scaled to your chosen mode’s size — there is no framing control — and stored under `~/ .config/sdroxide-brown/sstv_tx/`. **Clear** empties a slot’s picture but keeps its message.
- Type a **message** for the active slot. The **first line is drawn at double size as a title**, and a live preview shows exactly what goes out.
- **Banner...** sets the strip across the top of every picture you send. It understands {call}, {grid} and {version}, and you can style it: colours, a gradient, a text outline, a rainbow override, and its height.

- Choose a mode, or leave **Auto** (which sends **Martin 1** until it has detected one). There are 16, from 320×256 to 800×616. **PD120** and **PD180** give a 640×496 picture for about the same air time as the smaller modes — worth using for a nicer image.

C4. Pressing TX

TX composes the picture and keys the radio. **ABORT TX** stops one in progress. Three trims sit on the receive side, and all three earn their place:

- **FSK ID** (on by default) sends your callsign as tones after the picture, the way MMSSTV does. It is how you are identified, and it costs about **two and a half seconds** on top of a transmission that has already taken a minute.
- **TX lead** (0–3000 ms) covers your rig’s key-up delay. A frame opens with about a second of leader and VIS code, and **a decoder that misses any of it shows no picture at all** — so if a WebSDR hears you but shows nothing, raise this first.
- **TX slant** (±5000 ppm) trims the transmit clock. A few Hz of error over a two-minute picture is a visible slant, and this is the correction.

A picture is slow, and the channel is shared. Roughly a minute for Robot 36, two for Martin 1 or PD120, four and a half for Scottie DX — and it occupies about **2.7 kHz** of the sideband. One station holds the frequency for that whole time. **Listen before you send**, and on the calling frequency keep it short and move aside so others can call.

The transmit rails still apply, exactly as for any mode: a refuse message names the reason — SWR, the radio’s own transmit range, another radio on the air, or the rig already keyed on its own PTT.

C5. The rules

- **CB is meant for sending *and* receiving, and in most countries it needs no licence** — the licence-free CEPT channels, at limited power. This fork treats 11 m as a full band.
- The **Allow transmit on 11 m (CB)** switch opens **11 m and nothing else** — the broadcast bands stay receive-only.
- Keep to your country’s channels, power and modes — ordinary CB courtesy. **Check the current rules where you are.**
- This fork is for the whole of CB — voice and the digital modes — alongside SWL and decoding. CB and amateur radio are neighbours on the same spectrum.

Part D — Keeping a record

- Received pictures live as **PNG** files under `~/ .config/sdroxide-brown/sstv_rx/`; the gallery is stored on the machine the radio is plugged into, so every screen sees the same collection. Right-click a thumbnail to delete it.
- **Profiles (Settings > Profiles)** save a whole working setup — dial and mode, gain, drive, the digital identity and the band stacks — under a name.
- The SSTV panel is the same in the **browser client**; decode and encode run in the server engine.

Help and links

- [README](#) — the full overview of the fork.
- [USER MANUAL.md](#) — the manual; §3.6 is SSTV in full.

- [Releases](#) — the latest build for every platform.

See you on 27.700! 73